LITERATURE REVIEW

Random matrices: Universality of ESDs and the circular law

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ANALYTICAL ABSTRACT

The empirical spectral distribution of an $n \times n$ matrix with independent entries has been believed since the 1950s to converge, after division by $\sqrt{n}$, to the uniform law on the unit disk, but proofs required a moment of order $2 + \varepsilon$. This paper removes that extra moment. It proves a universality principle: if two matrices differ only in the law of their centred iid entries, both having unit variance, and the deterministic mean matrix has bounded normalised Hilbert-Schmidt norm, then their normalised spectral measures are asymptotically identical, in probability and, under a mild extra condition, almost surely. The proof replaces the eigenvalues by the log-determinant through Girko's Hermitization, controls the difference of log-determinants by splitting the singular values into a bulk handled by the Dozier-Silverstein theorem, a high-dimensional part handled by a Talagrand concentration bound on row distances, and a tail handled by polynomial least-singular-value bounds. Combined with Mehta's Ginibre computation, the circular law follows under second moments only. The review reconstructs each step, tests the claims with a simulation, and notes that the result is asymptotic and says nothing about the speed of convergence.

1 Introduction

1.1 Paper identification and classification

The paper is a theoretical contribution in probability theory, with the subfield of random matrix theory and, within it, the spectral theory of non-Hermitian ensembles. It belongs to the Pure Mathematics series because every result is a theorem proved from first principles, with no data other than illustrative simulations. The reviewed version is the fifth arXiv version of April 2009, which carries a short appendix by Krishnapur. The published version is Tao, Vu and Krishnapur (2010). The full text, including both appendices and the reference list, was read and reconstructed.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the limiting shape of the eigenvalue cloud of a large square matrix with independent, identically distributed entries depends on anything except the mean and variance of those entries. For Hermitian matrices the answer is known: only the first two moments matter (Wigner's semicircle law). For non-Hermitian matrices the corresponding statement, the circular law, had been established only under extra moment assumptions.

WHY DOES THE QUESTION MATTER?

Eigenvalues of non-normal matrices are unstable objects. They are not controlled by singular values, they have no variational characterisation, and none of the standard tools (moment method, Stieltjes transform of a self-adjoint operator) apply directly. A universality theorem that survives this difficulty gives a template for every model where an explicit eigenvalue density is unavailable. Applications include stability of large linear systems (May 1972), asymmetric neural networks (Sommers, Crisanti, Sompolinsky and Stein 1988) and, in finance, lead-lag and transition matrices estimated from many assets.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Ginibre (1965) gave the joint eigenvalue density for complex Gaussian entries, and Mehta (1967) used it to prove the disk limit in that case. Edelman (1988) did the real Gaussian case. Girko (1984) and Bai (1997) introduced and developed the Hermitization method for general entries, and Pan and Zhou, Gotze and Tikhomirov, and Tao and Vu (2008) pushed the moment assumption down to $2 + \varepsilon$. The missing piece was the finite-variance case.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes a replacement principle: the log-determinant is the only statistic that needs to match between two ensembles. It then proves that, for every fixed complex z, the log-determinants

of $n^{-1/2}(M_n + X_n) - zI$ and $n^{-1/2}(M_n + Y_n) - zI$ differ by $o(1)$. The consequences are a universality theorem, the circular law for any iid entries of mean zero and unit variance, an equivalence between convergence of eigenvalues and convergence of singular values of shifted matrices, and extensions to random and non-identically distributed models.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Let X_n and Y_n be $n \times n$ matrices whose entries are iid copies of complex random variables x and y with mean zero and variance one, and let M_n be deterministic with $\sup_n n^{-2}\|M_n\|_2^2 < \infty$. Does the difference between the empirical spectral distributions of $n^{-1/2}(M_n + X_n)$ and $n^{-1/2}(M_n + Y_n)$ converge to zero, and can this be proved with no moment assumption beyond the second? A corollary question is whether the circular law holds under exactly the second moment.

2.2 Hypotheses

- The limiting spectral measure of an iid matrix depends on the entry law only through the mean and variance (universality).
- Convergence of the eigenvalue measure is equivalent to convergence of the singular value measures of the shifted matrices $n^{-1/2}A_n - zI$ for almost every z, together with a uniform integrability condition on log.
- A lower-tail distance bound for a random row against a subspace of large codimension can replace the rate-of-convergence estimates for Hermitian ESDs that earlier proofs required.

2.3 Existing literature and research gap

PAPER The authors cite Wigner (1958) and Pastur (1973) for the Hermitian case with second moments only, Mehta (1967) and Edelman (1988) for Gaussian non-Hermitian matrices, and Girko (1984), Bai (1997), Gotze and Tikhomirov, Pan and Zhou and Tao and Vu (2008) for general entries under extra moment conditions. The gap is the extra ε.

CRITIQUE The framing is accurate. The deeper gap, which the paper identifies in Appendix C remark 3, is methodological: all earlier proofs after Bai needed a rate of convergence for the ESD of the Hermitized matrix $(A_n - zI)(A_n - zI)^*$. Rates are not available under second moments, so the contribution is to show that rates are unnecessary.

2.4 Claimed contribution

CRITIQUE Three contributions are claimed: the full circular law, the universality principle with a deterministic shift, and the equivalence theorem. All three are demonstrated by proof, subject to external inputs taken from other papers (the least singular value bound, the Dozier-Silverstein theorem, Talagrand's inequality). The proof is therefore complete modulo those cited results, which are standard and peer reviewed. One claim is conditional: almost sure universality requires that the ESDs of $(n^{-1/2}M_n - zI)(n^{-1/2}M_n - zI)^*$ converge, and the paper does not claim more.

3 Theoretical Framework and Mechanism

3.1 Objects and the main theorem

PAPER For $A \in M_n(\mathbb{C})$ with eigenvalues $\lambda_1, \dots, \lambda_n$ the empirical spectral distribution is the probability measure

$$\mu_A(s,t) = \frac{1}{n}\left|\{\, 1 \le i \le n : \ \mathrm{Re}\,\lambda_i \le s,\ \mathrm{Im}\,\lambda_i \le t \,\}\right|. \tag{1}$$

μ_A Discrete measure on $\mathbb{C}$ giving mass $1/n$ to each eigenvalue (with multiplicity).

$n^{-1/2}$ Normalisation. With iid entries of unit variance the eigenvalues are of size $\sqrt{n}$, so $n^{-1/2}A_n$ has a spectrum of order one.

Convergence is in the vague topology: $\mu_n \to \mu$ if $\int f d\mu_n \to \int f d\mu$ for every continuous compactly supported f, either in probability or almost surely.

> **THEOREM 1.7 (UNIVERSALITY PRINCIPLE)**
>
> Let x, y have mean zero and variance one, let X_n, Y_n have iid entries distributed as x, y, and let M_n satisfy
>
> $$\sup_n \frac{1}{n^2}\|M_n\|_2^2 < \infty, \tag{2}$$
>
> where $\|A\|_2 = \mathrm{trace}(AA^*)^{1/2}$. Put $A_n = M_n + X_n$, $B_n = M_n + Y_n$. Then $\mu_{n^{-1/2}A_n} - \mu_{n^{-1/2}B_n} \to 0$ in probability. If in addition the ESDs of $(n^{-1/2}M_n - zI)(n^{-1/2}M_n - zI)^*$ converge for almost every z, the convergence is almost sure.

Condition (2) says that the average squared entry of M_n is bounded. It is exactly the condition that makes the mean part comparable in size to the fluctuation part X_n, whose squared Hilbert-Schmidt norm is $\approx n^2$ by the law of large numbers. Lemma 1.9 records the consequence: $n^{-2}\|A_n\|_2^2$ and

$\int |z|^2 d\mu_{n^{-1/2}A_n}$ are almost surely bounded. The second claim uses Weyl's inequality (Lemma A.2),

$$\sum_{j=1}^{n} |\lambda_j|^2 \le \sum_{j=1}^{n} \sigma_j(A)^2 = \|A\|_2^2 = \sum_{i,j} |a_{ij}|^2, \tag{3}$$

proved by triangularising $A = QVQ^{-1}$ (Schur form), where the eigenvalues are the diagonal entries of the upper triangular V and $\|V\|_2 \ge$ the norm of its diagonal. Equation (3) is the reason second moments are the natural condition: it converts a bound on entries into tightness of the eigenvalue measure, with $\mu\{|z| \ge R\} \le C/R^2$.

3.2 Why the determinant: Girko's Hermitization

PAPER The eigenvalue measure of a non-normal matrix cannot be recovered from singular values of A alone. The route used here is the logarithmic potential. Because $\det(n^{-1/2}A_n - zI) = \prod_j (n^{-1/2}\lambda_j - z)$,

$$\frac{1}{n} \log\left|\det(n^{-1/2}A_n - zI)\right| = \int_{\mathbb{C}} \log|w - z| \, d\mu_{n^{-1/2}A_n}(w). \tag{4}$$

The right side is the logarithmic potential of the ESD. The identity $|\det A| = \prod_i \sigma_i(A) = \prod_i d_i(A)$ (equation 11 of the paper), where d_i is the distance from the ith row to the span of the previous rows, then expresses the same quantity through singular values or through row distances. Both are quantities one can estimate for a random matrix.

To invert (4) the paper defines the Stieltjes-like transform

$$g_n(z) = \frac{2}{n} \operatorname{Re} \sum_{j=1}^{n} \frac{z - n^{-1/2}\lambda_j}{|z - n^{-1/2}\lambda_j|^2} = \frac{\partial}{\partial \operatorname{Re} z}\left[\frac{2}{n} \log|\det(n^{-1/2}A_n - zI)|\right], \tag{5}$$

and Girko's identity (Lemma 3.3): for $u, v \neq 0$,

$$m_n(u,v) := \int e^{iu \operatorname{Re} w + iv \operatorname{Im} w} d\mu_{n^{-1/2}A_n}(w) = \frac{u^2+v^2}{4\pi i u} \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} g_n(s+it)\, e^{ius+ivt}\, dt \Big) ds. \tag{6}$$

$m_n(u,v)$ Characteristic function (Fourier transform) of the ESD.

g_n Real part of the Cauchy transform; equals $\partial_{\operatorname{Re} z}$ of twice the log-potential.

EVIDENCE The identity is a pointwise statement for each eigenvalue w, and it can be checked directly. The paper's contour computation (equation 19) gives, for $v > 0$ and $s \neq \operatorname{Re} w$,

$$\int_{\mathbb{R}} \frac{\operatorname{Re}(s+it-w)}{|s+it-w|^2} e^{ius+ivt} dt = \pi \operatorname{sgn}(s - \operatorname{Re} w)\, e^{-v|s-\operatorname{Re} w|}\, e^{ius} e^{iv \operatorname{Im} w}.$$

Writing $\sigma = s - \operatorname{Re} w$ and integrating in s,

$$\int_{\mathbb{R}} \operatorname{sgn}(\sigma) e^{-v|\sigma|} e^{iu\sigma} d\sigma = \frac{1}{v - iu} - \frac{1}{v + iu} = \frac{2iu}{u^2+v^2},$$

so the double integral equals $\frac{2\pi i u}{u^2+v^2} e^{i(u\,\mathrm{Re}\,w + v\,\mathrm{Im}\,w)}$, which rearranges to (6) for a single eigenvalue. Averaging over eigenvalues gives the lemma. For $v < 0$ the exponent takes $|v|$ and an extra sign appears, which the paper suppresses.

An alternative derivation due to Krishnapur (Section 3.6) uses the fact that the distributional Laplacian of $\log|P|$ is the counting measure of the zeros of a polynomial P. By Green's theorem, for smooth compactly supported f,

$$\int f d\big(\mu_{n^{-1/2}A_n} - \mu_{n^{-1/2}B_n}\big) = \frac{1}{2\pi n} \int_{\mathbb{C}} \Delta f(z) \Big[\log|\det(n^{-1/2}A_n - zI)| - \log|\det(n^{-1/2}B_n - zI)|\Big] dz,$$

which makes the logic transparent: the difference of ESDs is a smoothed difference of log-determinants.

3.3 The replacement principle

> **THEOREM 2.1 (REPLACEMENT PRINCIPLE)**
>
> Let $A_n, B_n \in M_n(\mathbb{C})$ be random matrices, not necessarily independent. Suppose (i) $n^{-2}\|A_n\|_2^2 + n^{-2}\|B_n\|_2^2$ is bounded in probability (resp. almost surely), and (ii) for almost every z,
>
> $$\frac{1}{n}\log|\det(n^{-1/2}A_n - zI)| - \frac{1}{n}\log|\det(n^{-1/2}B_n - zI)| \to 0$$
>
> in probability (resp. almost surely). Then $\mu_{n^{-1/2}A_n} - \mu_{n^{-1/2}B_n} \to 0$ in the same mode.

PAPER The proof is a dominated convergence argument. Lemma 3.1 states that if $f_n : X \to \mathbb{R}$ satisfy a uniform bound $\int |f_n|^{1+\delta} d\nu = O_P(1)$ on a finite measure space and $f_n(x) \to 0$ pointwise in probability for ν-a.e. x, then $\int f_n \, d\nu \to 0$ in probability. It is applied with $f_n(s,t) = \frac{1}{n}\log|\det(\cdot)| - \frac{1}{n}\log|\det(\cdot)|$ after integrating (6) by parts in s and localising with a cutoff $\psi(\cdot/R^2)$. The uniform integrability (27) comes from (3): $\int \log^2|n^{-1/2}\lambda - (s+it)|\,|\phi_{u,v,R}| \le O(1+|\lambda|^2/n)$, so the L^2 norm of f_n is controlled by $n^{-2}\|A_n\|_2^2$. The theorem is "soft": no independence is assumed. All the probabilistic content is therefore concentrated in condition (ii).

3.4 The determinant estimate

PAPER Proposition 2.2 delivers (ii). After shifting M_n by $\sqrt{n}zI$ one may take $z = 0$ and must show $\frac{1}{n}\log|\det n^{-1/2}A_n| - \frac{1}{n}\log|\det n^{-1/2}B_n| \to 0$. Let X_i be the rows of A_n, $V_i = \mathrm{span}(X_1, \dots, X_{i-1})$. Then

$$\frac{1}{n}\log\big|\det(n^{-1/2}A_n)\big| = \frac{1}{n}\sum_{i=1}^{n} \log \mathrm{dist}(n^{-1/2}X_i, V_i). \tag{7}$$

The sum is split into three ranges of i, and the argument is the heart of the paper.

STEP 1: THE LAST $n^{0.99}$ ROWS (LEMMA 4.1)

A theorem of Tao and Vu (2008, 2010) on the least singular value, plus Borel-Cantelli, gives $\sigma_n(A_n), \sigma_n(B_n) \geq n^{-O(1)}$ almost surely for large n, while the crude upper bound $\sigma_1 \leq n^{O(1)}$ holds because $\|M_n\| = O(n)$. By Lemma A.4 each logarithm is $O(\log n)$, so these $n^{0.99}$ terms contribute at most $O(n^{0.99} \log n/n) = o(1)$.

STEP 2: THE HIGH-DIMENSIONAL ROWS, $(1-\delta)n \leq i \leq n - n^{0.99}$ (LEMMA 4.2)

Here no matching between A_n and B_n is attempted. Instead each log is shown to be small in absolute value. The upper part $\max(\log \mathrm{dist}, 0)$ is bounded by Chebyshev and Hoeffding. The lower part needs the lower-tail estimate below, which is the technical novelty.

STEP 3: THE BULK, $1 \leq i \leq (1-\delta)n$ (LEMMA 4.3)

The product of distances is the volume of the parallelepiped on the first $n' = \lfloor (1-\delta)n \rfloor$ rows, equal to $\det(n^{-1} A_{n,n'} A_{n,n'}^*)^{1/2} = \prod_i n^{-1/2} \sigma_i(A_{n,n'})$. The target becomes $\int_0^\infty \log t \, d\nu_{n,n'}(t) = O(\varepsilon)$ for the difference ν of two singular value ESDs, which is handled by comparing with the Dozier-Silverstein theorem after cutting t into four ranges.

3.5 The lower-tail bound and its proof

PAPER Proposition 5.1 states that for a deterministic d-dimensional subspace $W \subset \mathbb{C}^n$ with $d \leq n - n^{0.99}$, a row X of A_n and any $0 < c < 1$,

$$\mathbf{P}\big(\mathrm{dist}(X, W) \leq c\sqrt{n-d}\big) = O\big(\exp(-n^{0.01})\big). \tag{8}$$

The reduction has four moves. First, the mean $v = \mathbf{E}X$ is absorbed: $\mathrm{dist}(X, W) \geq \mathrm{dist}(X', \mathrm{span}(W, v))$ with X' centred, which raises d by one. Second, since $\mathbf{P}(|x_i| \geq n^{0.1}) = O(n^{-0.2})$ by Chebyshev, a Chernoff bound shows that with probability $1 - O(\exp(-n^{0.01}))$ at most $n^{0.9}$ coordinates exceed $n^{0.1}$; conditioning on their positions and projecting onto the remaining coordinates reduces to bounded entries. Third, the truncated variable $\tilde{x} = x \,|\, |x| < n^{0.1}$ has variance $1 - o(1)$. Fourth, two estimates are combined.

PAPER *Concentration.* Talagrand's inequality (Theorem 5.2): for a product probability on $\mathbf{D}^n$ ($\mathbf{D}$ the unit disk), a convex 1-Lipschitz F and $r \geq 0$,

$$\mu\big(|F - M(F)| \geq r\big) \leq 4 \exp(-r^2/8), \tag{9}$$

with $M(F)$ the median. Apply it to $F(v) = \mathrm{dist}(v, W)$, which is convex and 1-Lipschitz, with μ the law of $\tilde{X}/n^{0.1}$:

$$\mathbf{P}\Big(|\,\mathrm{dist}(\tilde{X}, W) - M| \geq n^{0.1} r\Big) \leq 4e^{-r^2/8}.$$

PAPER *Second moment* (Lemma 5.3). With π the orthogonal projection whose range has dimension $n-d$ (the printed text calls it the projection onto W, but the trace $\tilde{n}$ it uses is that of the projection onto

$W^{\perp}$, of rank $n-d$),

$$\operatorname{dist}(\tilde{X}, W)^2 = \sum_{i,j} \tilde{x}_i \pi_{ij} \overline{\tilde{x}_j}, \qquad \mathbf{E}\operatorname{dist}(\tilde{X}, W)^2 = \mathbf{E}|\tilde{x}|^2 \operatorname{trace}\pi = (1-o(1))(n-d),$$

using independence and zero mean to kill off-diagonal terms.

INFERENCE Combining the two gives (8) with room to spare. The median satisfies $M^2 = (1-o(1))(n-d)$ because the fluctuation of dist is $O(n^{0.1})$ by (9) whereas $\sqrt{n-d} \geq n^{0.495}$. For $c < 1$ the event dist $\leq c\sqrt{n-d}$ then needs a deviation of at least $(1-c-o(1))\sqrt{n-d} \geq c' n^{0.495}$, so $r \geq c' n^{0.395}$ and the probability is at most $4\exp(-c'' n^{0.79})$, far below $\exp(-n^{0.01})$. The paper's exponent is thus conservative, which is harmless since it only needs a summable bound for Borel-Cantelli.

INFERENCE The consequence for the sum (7) can be computed explicitly. For i in the range, (8) with $c = 1/2$ gives $\operatorname{dist}(n^{-1/2}X_i, V_i) \geq \frac{1}{2}\sqrt{(n-i+1)/n}$, so with $k = n-i+1$,

$$\frac{1}{n}\sum_{k=n^{0.99}}^{\delta n}\Big(-\log \operatorname{dist}\Big)^{+} \leq \frac{1}{n}\sum_{k=1}^{\delta n}\Big[\tfrac{1}{2}\log\frac{n}{k} + \log 2\Big] \approx \tfrac{1}{2}\int_0^{\delta} \log\frac{1}{x}\,dx + \delta\log 2 = \tfrac{1}{2}\delta\big(1+\log\tfrac{1}{\delta}\big) + \delta\log 2,$$

which tends to 0 as $\delta \to 0$. This is the quantitative content of Lemma 4.2: the last δn rows are nearly orthogonal enough that their log-contribution vanishes with δ.

3.6 The bulk: matching singular values

PAPER For the range $\delta^2 \leq t \leq \varepsilon^4$ and $t < \delta^2$ the paper needs a lower bound on the singular values of the truncated matrix $A_{n,n'}$. The negative second moment identity is exact (Lemma A.4): for a full rank $n' \times n$ matrix with rows X_j and $W_j = \operatorname{span}(X_i : i \neq j)$,

$$\sum_{j=1}^{n'} \sigma_j(A)^{-2} = \sum_{j=1}^{n'} \operatorname{dist}(X_j, W_j)^{-2}. \tag{10}$$

EVIDENCE Proof: $\operatorname{trace}(AA^*)^{-1} = \sum_j \sigma_j^{-2}$. Put $v_j = (AA^*)^{-1}e_j$. Then A^*v_j is orthogonal to every $X_i = A^*e_i$ with $i \neq j$ (so to W_j) and has inner product 1 with X_j, whence $v_{jj}\operatorname{dist}(X_j, W_j)^2 = 1$ and $\sum_j v_{jj} = \sum_j \operatorname{dist}(X_j, W_j)^{-2}$. Combined with (8) and Borel-Cantelli, this yields $n^{-1}\sum_i (n^{-1/2}\sigma_i)^{-2} = O(1)$ (Lemma 6.6), and, via Cauchy interlacing $\sigma_i(A) \geq \sigma_i(A') \geq \sigma_{i+k}(A)$ (Lemma A.1),

$$n^{-1/2}\sigma_i(A_{n,n'}) \geq c\,\frac{n'-i}{n} \qquad (1 \leq i \leq (1-2\delta)n) \tag{11}$$

almost surely for large n (Lemma 6.7). Equation (11) says that the singular values near the bottom of the spectrum are repelled from zero linearly. It is exactly the information needed to control $\int |\log t|\,d\nu$ near $t = 0$ because $|\log t| \leq \varepsilon t^{-2}$ there and the negative second moment is bounded.

PAPER For intermediate t, with ψ a smooth cutoff equal to one on $[\varepsilon^4, R_\varepsilon]$, interlacing allows replacing n' by n at cost $O(\varepsilon)$, and then the following theorem gives that the singular value ESDs of A_n and B_n converge to the same limit.

THEOREM B.1 (DOZIER AND SILVERSTEIN 2007)

Let X_n be $n \times r$ with iid entries of variance one, $r = (c + o(1))n$, and M_n a random $n \times r$ matrix independent of X_n whose $M_n M_n^*$ has ESD converging to H. Then the ESD of $C_n = \frac{c}{n}(M_n + X_n)(M_n + X_n)^*$ converges almost surely to a measure F whose Stieltjes transform $m(\lambda) = \int (\lambda' - \lambda)^{-1} dF(\lambda')$ solves

$$m = \int \frac{dH(t)}{\frac{t}{1+cm} - (1 + cm)\lambda + (1 - c)}. \tag{12}$$

EVIDENCE Only H enters (12); the law of x does not. This is the "universality of the singular values" that the Hermitized problem requires. As a check, take $M_n = 0$, so $H = \delta_0$, and $c = 1$: then $m = \left[-(1+m)\lambda\right]^{-1}$, that is $\lambda m^2 + \lambda m + 1 = 0$, the Marchenko-Pastur law with ratio one. Thus the novelty is not this step but the removal of any need for its rate.

3.7 Equivalence theorem

THEOREM 1.20 (EQUIVALENCES FOR CONVERGENCE)

Let μ be a probability measure on $\mathbb{C}$ with $\int |z|^2 d\mu < \infty$. The following are equivalent, in probability, and almost surely if the ESDs (4) converge: (i) $\mu_{n^{-1/2}A_n} \to \mu$; (ii) for almost every z, $\frac{1}{n} \log |\det(n^{-1/2}A_n - zI)| \to \int \log|w - z|\, d\mu(w)$; (iii) for almost every z and some $\varepsilon_n \downarrow 0$, $\frac{1}{n} \log\det((n^{-1/2}A_n - zI + \varepsilon_n I)(n^{-1/2}A_n - zI)^* + \varepsilon_n I) \to 2\int \log|w - z|\, d\mu(w)$.

PAPER (i) $\Leftrightarrow$ (ii) uses (4): the log-determinant is the integral of $\log|\cdot - z|$ against the ESD. Convergence against a function unbounded at both ends needs uniform integrability, and the two bounds supplied are (3) at infinity and the singular value lower bound (11) at the singularity. (ii) $\Leftrightarrow$ (iii) states that the log-determinant is stable to a regularisation ε_n that is slow enough, again by (11). The regularised object in (iii) is a Hermitian determinant, so its limit is accessible by Stieltjes transform methods. The theorem therefore reduces the eigenvalue problem to the singular value problem for the family $\{n^{-1/2}A_n - zI\}_z$.

INFERENCE As an illustration, for the circular law μ uniform on the unit disk the right side of (ii) is the logarithmic potential

$$U(z) = \int \log|w - z|\, d\mu(w) = \begin{cases} \frac{1}{2}(|z|^2 - 1), & |z| \le 1, \\ \log|z|, & |z| > 1, \end{cases}$$

which satisfies $\Delta U = 2\pi \times$ density on the disk and is checked at $z = 0$ by $\int_0^1 2r \log r\, dr = -\frac{1}{2}$. This is used below to test the theory numerically.

3.8 Extensions

PAPER Corollary 1.15: if M_n has rank $o(n)$ and obeys (2), the limit is still the disk, because a low-rank self-adjoint perturbation of $|z|^2 I$ leaves the ESD of $(n^{-1/2}M_n - zI)(\cdot)^*$ concentrated at $|z|^2$. Theorem 1.17 allows a random M_n independent of the entries by conditioning. Theorem 1.18 treats $A_n = M_n + K_n X_n L_n$, with the key identity

$$\det\bigl(n^{-1/2}A_n - zI\bigr) = \det\Bigl(n^{-1/2}(K_n^{-1}M_nL_n^{-1} + X_n) - zK_n^{-1}L_n^{-1}\Bigr)\det(L_nK_n),$$

in which $\det(L_nK_n)$ cancels in the difference of logarithms. Appendix C (Krishnapur) treats independent but not identically distributed entries $a_{ij} = \mu_{ij} + \sigma_{ij}x_{ij}$ with $a \le \sigma_{ij} \le b$, replacing Theorem B.1 by Chatterjee's invariance principle for Hermitian matrices, at the price of weak convergence only.

4 Data and Empirical Strategy

EVIDENCE The paper contains no empirical data. The only numerical content is three figures of eigenvalue plots of 5000×5000 matrices (Bernoulli ± 1 and Gaussian entries; shifted diagonal M_n; and $A + BM_nB$ forms), produced by P. Matchett Wood, which illustrate the circular law, Corollary 1.10 and Theorem 1.18. The identification question for a theoretical paper is whether the stated conditions imply the stated conclusion. The proof strategy is the identification.

4.1 Reviewer simulation (generated, not from the paper)

The reviewer simulated three entry laws, each centred with unit variance, and computed eigenvalues of $n^{-1/2}X_n$: standard Gaussian; Bernoulli ± 1; and a symmetric heavy-tailed law defined by $\mathbf{P}(|x| > t) = e^2/(t^2 \log^2 t)$ for $t \ge e$, scaled by $(\sqrt{3}\,e)^{-1}$. The last law has $\mathbf{E}x^2 = \int_0^\infty 2t\,\mathbf{P}(|x| > t)dt = e^2 + 2e^2\int_e^\infty \frac{dt}{t\log^2 t} = 3e^2$, so the variance is finite, but $\mathbf{E}|x|^{2+\varepsilon} = \infty$ for every $\varepsilon > 0$. It is therefore a case covered by this paper and by no earlier one. The random seed is fixed (20261101); the numbers below are means over replications and are an estimate with the stated standard deviation, not an official source.

Table 1. Simulation design and statistics reported

Quantity	Construction	Sample	Replications
Eigenvalue cloud	eigenvalues of $n^{-1/2}X_n$	$n = 1500$	1
Sup distance of radial CDF	$\sup_r \lvert F_n(r) - \min(r,1)^2\rvert$	$n \in \{100, \dots, 1200\}$	6 per n
Mean log-determinant	$\frac{1}{n}\log\lvert\det(n^{-1/2}X_n - zI)\rvert$	$n = 1000$, $z \in \{0, 0.5, 1.5\}$	5 per z

Source: generated by the reviewer in Python (NumPy); entries drawn by inverse transform for the heavy-tailed law; not from the paper.

4.2 Empirical specification and identification

The identifying variation in a theoretical paper is the randomness of the entries and the structure of the proof. Two features matter. First, the conclusion is about the difference of the two ESDs, so the unknown limit never has to be computed; the Gaussian case is borrowed from Mehta (1967). Second, only the mean and the variance of the entry law enter, so the proof must show that every higher-order feature of x is washed out. This happens because the log-determinant is a sum of n weakly dependent terms log dist, each concentrated, and the terms where concentration fails (the last δn rows) have vanishing weight.

4.3 Data limitations and biases

- The simulation uses finite $n \leq 1500$, while the theorem is asymptotic; it can only illustrate the rate, not test the limit.
- The heavy-tailed law is one specific example; no claim is made about every law with $\mathbf{E}|x|^2 = 1$.
- Eigenvalues of non-normal matrices are ill-conditioned, so numerical eigenvalues carry rounding error, small compared with the effect studied here.
- The paper's figures show only $n = 5000$ single realisations, with no quantitative metric.

5 Results

5.1 Main theorems

PAPER The main results are Theorem 1.7 (universality), Theorem 1.13 (circular law), Corollary 1.15 (shifted circular law for M_n of rank $o(n)$), Theorem 1.20 (equivalences) and Theorem 1.23 (existence of a limit). Theorem 1.13 follows from Corollary 1.10 and Mehta: with $M_n = 0$ and y complex Gaussian, $\mu_{n^{-1/2}Y_n}$ is the Ginibre ensemble, whose joint density

$$p(\lambda_1, \dots, \lambda_n) = c_n \prod_{i<j} |\lambda_i - \lambda_j|^2 \exp\Big(-n \sum_{i=1}^{n} |\lambda_i|^2\Big) \tag{13}$$

gives the uniform law on the disk. The factor $\prod |\lambda_i - \lambda_j|^2$ is a Coulomb repulsion between eigenvalues, the factor $\exp(-n \sum |\lambda_i|^2)$ a confining harmonic potential; balancing the two yields constant density $1/\pi$ inside the unit disk. The theorem thus says that this two-dimensional Coulomb gas is the universal description of all iid matrices at the level of density.

EVIDENCE Magnitude: these are qualitative limit statements, so there is no coefficient. What can be quantified is how far a finite matrix is from the limit. In the reviewer's simulation (Table 2 and Figure 2) the sup-distance between the empirical radial distribution and the limit r^2 is about 0.068 at $n = 100$ and 0.014 at $n = 1200$ for Gaussian entries, a decline of roughly a factor 4.9 for a twelve-fold increase in n, that is a decay of order $n^{-0.6}$ on this range.

5.2 Numerical illustration of the universality claim

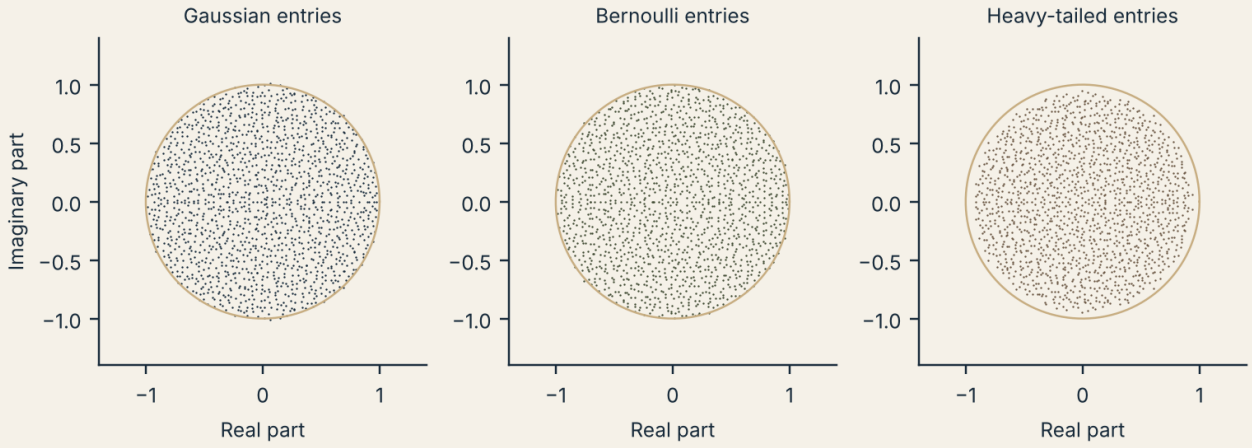
Table 2. **Reviewer simulation: normalised log-determinant versus the logarithmic potential of the unit disk, $n = 1000$**

Shift z	Theory $U(z)$	Gaussian	Bernoulli	Heavy-tailed
0	−0.5000	−0.5022	−0.4990	−0.5768
0.5	−0.3750	−0.3793	−0.3757	−0.4274
1.5	0.4055	0.4050	0.4050	0.4045

Source: generated by the reviewer. Entries are means over five matrices; standard deviations are below 0.0035 for Gaussian and Bernoulli and 0.005 for the heavy-tailed law. Theory column is $\frac{1}{2}(|z|^2 - 1)$ for $|z| \le 1$ and $\log |z|$ for $|z| > 1$ (analytic).

EVIDENCE Table 2 answers a narrow question: does the statistic the proof uses, the normalised log-determinant of Equation (4), converge to the log-potential of the disk? The dependent variable is $\frac{1}{n} \log |\det|$, the independent variable is the shift z (inside, boundary region, outside the disk). The Gaussian and Bernoulli columns agree with theory to within about 0.004, and with each other, which is the universality statement. Outside the disk ($z = 1.5$) all three laws agree to 0.001, because there the determinant is dominated by deterministic growth of $|z|$ and is not sensitive to the entry law. Inside the disk the heavy-tailed column is lower by 0.077 at $z = 0$ and 0.052 at $z = 0.5$. The table does not show that the heavy-tailed law fails in the limit, only that at $n = 1000$ it is far from it; the theorem predicts convergence, and the simulation shows how slowly it is approached.

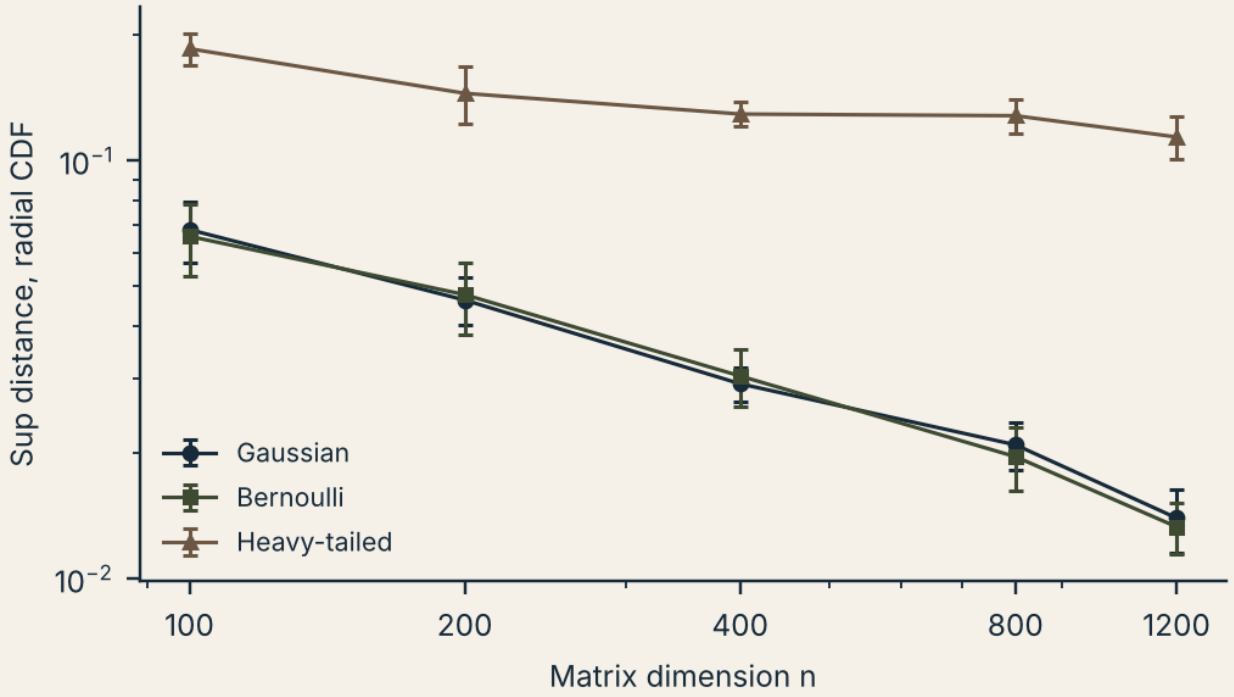
Figure 1. **Eigenvalues of $n^{-1/2} X_n$ for three entry laws, $n = 1500$, one realisation each**



Source: generated by the reviewer. The circle is the unit circle. Entries have mean zero and variance one in all panels; the heavy-tailed law has $\mathbf{P}(|x| > t) = e^2/(t^2 \log^2 t)$, $t \ge e$, scaled to unit variance.

EVIDENCE Figure 2 plots the sup-distance between the empirical distribution of $|\lambda|$ and the limit $F(r) = \min(r, 1)^2$ against n on logarithmic axes. The Gaussian and Bernoulli curves are indistinguishable within one standard deviation at every n, which is the visible form of Theorem 1.7. The heavy-tailed curve falls from 0.185 to 0.114 over the same range and is still roughly an order of magnitude above the others at $n = 1200$. In the third panel of Figure 1 the eigenvalues occupy visibly less than the full disk, with the outer annulus empty. Nothing in the figure shows non-convergence; it shows convergence on a scale much larger than $n = 1200$.

Figure 2. **Sup distance between empirical and limiting radial distribution, against matrix dimension**



Source: generated by the reviewer. Means over six replications; error bars are one standard deviation. Both axes logarithmic. Limit radial CDF is r^2 on $[0, 1]$.

5.3 Economic significance

CRITIQUE The paper makes no economic claim. The relevant significance is methodological: it justifies using the disk as a null benchmark for the dynamics of large systems under assumptions that are far weaker than Gaussianity or bounded moments. For financial returns, which are commonly modelled with tail exponents between two and four, the paper shows that the null is asymptotically valid, but the simulation indicates that finite-sample accuracy may be very poor for tails that approach the variance boundary.

6 Robustness and Additional Tests

CRITIQUE A theoretical paper has no robustness tests in the empirical sense. The analogue is the set of generalisations and the places where the proof can be stress-tested. The table reviews each.

Table 3. **Robustness analogues: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Non-zero mean of the entries	Corollary 1.15 and the mean-subtraction in Proposition 5.1	Disk limit persists if the mean matrix has rank $o(n)$ and obeys (2)	Yes for low-rank means. Not for full-rank means, where the support changes (Remark 1.14).
Unbounded entries	Truncation at $n^{0.1}$ with Chernoff count of exceptions	Reduces to bounded entries up to probability $e^{-n^{0.01}}$	Yes, rigorous and standard.
Dependence on a rate for Hermitian ESDs	Replacement principle plus Lemma 4.2	No rate required	Yes. This is the main technical point.
Almost sure versus in probability	Theorem 1.7 second clause	Almost sure needs convergence of ESDs (4)	Partly. A real restriction on M_n, removed by passing to subsequences only for the in-probability statement.
Random or structured matrices	Theorems 1.17 and 1.18, Appendix C	Extends under bounds (5) and controlled second moment	Partly. Conditions on M_n, K_n, L_n are not verified for any example.
Independent but not iid entries	Appendix C	Weak convergence only, with κ-controlled second moment and Pastur condition	Partly. Almost sure convergence is left open.
Sparse matrices	Remark in Section 1.16	Expected to hold; not proved	No. Stated as an expectation only.

Source: reviewer's assessment of the paper.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

The circular law holds, in probability and almost surely, for iid complex entries with mean zero and variance one, with no further moment assumption. Universality holds for the shifted ensemble $M_n + X_n$ under (2). The equivalence between convergence of eigenvalues and convergence of singular values of shifted matrices is exact. These are proved theorems, conditional only on the cited external results.

SUPPORTED BUT CONDITIONAL

Almost sure universality needs convergence of the singular value ESDs of $n^{-1/2}M_n - zI$. Results for $A_n = M_n + K_n X_n L_n$ need the boundedness conditions (5) and (6), which are not verified for named examples. The existence of a limit (Theorem 1.23) is conditional on convergence of the shifted singular value distributions.

SUGGESTIVE

Universality for sparse iid matrices, for matrices with independent but not identically distributed entries almost surely, and for general dependence structures. The authors remark that the tools extend; they do not prove it.

NOT ESTABLISHED

Any statement about the speed of convergence, the behaviour of the largest eigenvalue, local statistics at the edge or in the bulk, or the behaviour of individual eigenvalues. The theorems concern the weak limit of the empirical measure only.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Removal of the ε: the circular law is proved in the optimal generality for iid entries, since variance one is necessary for the normalisation.
- Softness of the replacement principle: no independence is required between A_n and B_n, so the principle is reusable for other pairs of ensembles.
- Reduction of the non-Hermitian problem to Hermitian singular value laws, which gives a second proof route that avoids the explicit Ginibre formula.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Structural clarity: the proof isolates exactly where each ingredient is used (row distances for the top, Dozier-Silverstein for the bulk, least singular value for the last rows), so one can see which parts would break under a weaker assumption.
- Self-contained linear algebra (Appendix A): the identities (3) and (10) are short, verifiable, and the reader can reproduce them.
- The simulations in this review show that the headline universality (Gaussian versus Bernoulli) is numerically sharp already at $n = 100$, which supports the practical usefulness of the principle.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The result is purely asymptotic with no rate: $o(1)$ is never quantified. The proof uses Borel-Cantelli over events of probability $e^{-n^{0.01}}$, and the choice of δ in terms of ε is non-explicit. **Why it matters.** Every application involves finite n. For the heavy-tailed law with finite variance but no $2+\varepsilon$ moment, the reviewer's simulation shows a sup-distance of 0.114 at $n = 1200$, eight times that of the Gaussian case, and the log-determinant bias of 0.077 at $z = 0$. **Potential bias.** A user who invokes the circular law as a null for matrices of dimension 100 to 1000 with heavy-tailed entries will understate

the spectral radius excess and over-reject. **How it could be tested.** Prove a polynomial rate under an ε-moment and a sharper tail quantification, and in practice calibrate by Monte Carlo at the intended n and tail index.

OTHER WEAKNESSES

- Reliance on external heavy results. The least singular value bound (Tao and Vu 2008, 2010) is the deepest ingredient, and the paper uses it as a black box; the reader cannot verify the whole argument from this text. Potential bias: none in truth, but a gap in self-containedness. Test: reprove under the same assumptions.
- Minor typographical errors in the arXiv version. The projection in Lemma 5.3 is described as onto W rather than $W^{\perp}$. The Hoeffding bound in the proof of Lemma 4.2 is written with an exponent $-cn^{-0.01}$, which should read $-cn^{0.01}$ to be summable. The extension of Lemma 3.2 uses a stray notation $d\mu_{n^{-1/2}A_n}(u,v)$. None changes the argument, but they impede checking.
- Condition (2) is on $\|M_n\|_2$ only. It excludes means of order larger than constant per entry, which are common in applications where the mean matrix is a large drift or an adjacency matrix.
- The independence of X_n from M_n is used essentially (Theorem 1.17). Models where the mean depends on the noise, such as estimated coefficient matrices, are outside the theory.
- Talagrand's inequality is used in a complex-disk version with constant $1/8$ instead of $1/4$; the paper asserts that the original proof adapts but does not give it. Potential bias: none expected, but the step is unverified in the text.

8 Contradictions and Edge Cases

CRITIQUE *Where should the result disappear?* If the mean of the entries is non-zero and of full rank, the support is not the disk. A rank-one shift $M_n = \mu \mathbf{1}\mathbf{1}^{\top}$ has $\|M_n\|_2^2 = \mu^2 n^2$, so (2) holds, and the paper's Corollary 1.15 says the ESD of $n^{-1/2}A_n$ still tends to the disk. This is correct, and the explanation is that one eigenvalue sits at $\approx \mu\sqrt{n}$ after normalisation, an outlier of weight $1/n$ that vanishes under a vague limit. The result disappears as a statement about *spectral radius*: the largest eigenvalue modulus is not controlled by the ESD limit. In applications where the spectral radius is the object (stability), the theorem is silent.

CRITIQUE *Extreme values.* The theory requires $\mathbf{E}|x|^2 = 1$ and nothing more. For the heavy-tailed law used in the simulation, $\mathbf{P}(|x| > t) = e^2/(t^2 \log^2 t)$ before scaling, the largest of the n^2 entries of X_n is of order $n/\log n$, so the largest entry of $n^{-1/2}X_n$ is of order $\sqrt{n}/\log n \to \infty$ and the operator norm diverges. The ESD is nonetheless unaffected, because these entries touch $O(1)$ rows and columns out of n. The cost at finite n can be computed. Before scaling, the variance carried by entries not exceeding a level T is

$$\mathbf{E}\big[x^2 \mathbf{1}\{|x| \le T\}\big] = 3e^2 - \frac{2e^2}{\log T} - \frac{e^2}{\log^2 T}, \qquad \text{a fraction} \quad 1 - \frac{2}{3\log T} - \frac{1}{3\log^2 T} \text{ of } 3e^2.$$

Let T_n solve $n^2\,\mathbf{P}(|x| > T_n) = 1$, so that a typical matrix contains about one entry above T_n. For $n = 1000$, $T_n \approx 445$, $\log T_n \approx 6.10$, and the retained fraction of variance is $1 - 0.109 - 0.009 \approx 0.88$. A

typical realisation therefore behaves like a matrix of effective variance $s^2 \approx 0.88$, whose bulk has radius $s \approx 0.94$ and whose normalised log-determinant at $z = 0$ is $\log s - \frac{1}{2} \approx -0.564$. The reviewer's simulation gives -0.577 (Table 2), so the estimate accounts for most of the observed deficit of 0.077. This is a reviewer estimate, derived from the stated tail law and not from the paper. The missing variance fraction decays like $2/(3\log n)$, which is why convergence is so slow: halving the gap requires squaring n.

CRITIQUE *Parameter dependence.* The condition $\sup_n n^{-2}\|M_n\|_2^2 < \infty$ is sharp in scaling. If $M_n = n^{1/2+\eta}$ times a unitary, $\|M_n\|_2^2 = n^{2+2\eta}$ and the entries dominate the noise, and the ESD is that of M_n, a non-universal object. The theorem is silent beyond the stated threshold, and the answer changes continuously with the signal-to-noise ratio, in the manner described for the diagonal case by the support formula $\int |z-x|^{-2} d\mu(x) \geq 1$ cited in Remark 1.14.

CRITIQUE *Alternative mechanisms.* Universality could come from a central limit mechanism acting entry by entry rather than from the determinant structure. The paper's route shows that this is not the case: the Lindeberg replacement is applied to the *log-determinant*, a nonlinear global functional, and the heavy tail in the simulation (which retains the iid structure but loses the Gaussian-like concentration) exposes how the proof's concentration step (Talagrand on bounded entries) is what makes the limit appear.

CRITIQUE *Outside the setting.* The theorem needs independence of entries. Matrices of estimated lead-lag coefficients inherit cross-correlation from the common factor structure of returns, and the iid assumption fails. Appendix C allows independent non-identical entries but not dependence, so the theory has nothing to say for correlated columns, where the limit is governed by the population covariance.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Ginibre (1965); Mehta (1967)	Explicit eigenvalue density of the complex Gaussian ensemble and its disk limit	Foundation. The Gaussian case that this paper transfers to all laws.	Provides the only unconditional input from outside the proof.
Wigner (1958); Pastur (1973)	Semicircle law with second moments only	Hermitian precedent	Shows the same universality at the Hermitian level; the present paper is the non-Hermitian analogue.
Girko (1984); Bai (1997)	Hermitization and logarithmic potential; circular law under moment and density assumptions	Method preceded this paper	Source of the technique; Bai needed rates and bounded density.
Tao and Vu (2008)	Circular law under $2+\varepsilon$ moments, least singular value bound	Direct predecessor	Supplies the polynomial least singular value bound used here as a black box.

Paper	Shared idea	Relationship	Why it matters
Gotze and Tikhomirov (2010); Pan and Zhou (2010)	Circular law with weaker conditions by different routes	Parallel work	Independent confirmation of the Hermitization strategy.
Dozier and Silverstein (2007)	Limiting ESD of information-plus-noise matrices	Used inside the proof	Gives universality for the singular value bulk.
Chatterjee (2006)	Lindeberg-type invariance principle for functions of independent variables	Alternative tool, used in Appendix C	Allows independent non-identical entries at the cost of weak convergence.
Rudelson and Vershynin (2008)	Least singular value of a random square matrix at order $n^{-1/2}$	Improved the key input	Sharper than the polynomial bound used here.
Bordenave and Chafai (2012)	Survey of the circular law and its extensions	Later synthesis	Summarises what followed, including sparse and heavy-tailed generalisations.
May (1972); Sommers et al. (1988)	Stability of large random linear systems; spectrum of random asymmetric matrices	Application precedent	The circular law gives the stability threshold $\sigma\sqrt{nC} = 1$ and is the reason this paper matters outside mathematics.
Laloux et al. (1999); Plerou et al. (2002)	Marchenko-Pastur noise bands for correlation matrices in finance	Hermitian counterpart in the library	Same logic applied to symmetric empirical correlation matrices.

Source: reviewer's synthesis from the paper's reference list and standard literature; no citation beyond those listed in the bibliography.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** A rate of convergence for the ESD under finite variance. *Why it matters:* Practical use at finite n. *Existing limitation:* The proof is by contradiction and Borel-Cantelli, with no explicit dependence of n_0 on ε. *Way forward:* Track the dependence of δ on ε through Lemmas 4.2 and 4.3, or use sharper tail bounds under a $2 + \varepsilon$ moment.
- **Gap:** Almost sure universality without convergence of the shifted singular values. *Why it matters:* The extra hypothesis is not natural. *Existing limitation:* Theorem B.1 requires a limit for H. *Way forward:* A quantitative invariance principle for Hermitized matrices.

10.2 Literature-level gaps

- **Gap:** Universality for matrices with correlated entries and a factor structure. *Why it matters:* Financial and neural data are correlated. *Existing limitation:* Independence is used in every concentration step. *Way forward:* Replace Talagrand by concentration for dependent ensembles (for instance with a Lipschitz-in-ℓ^2 structure) and Hermitization with free probability.

– **Gap:** Behaviour of the spectral radius under finite variance. *Why it matters:* Stability analysis needs the largest eigenvalue, not the empirical measure. *Existing limitation:* The ESD does not control the extreme eigenvalue. *Way forward:* Extreme-value analysis of the log-determinant near the boundary.

10.3 Methodological gaps

– **Gap:** Non-asymptotic, finite-n bounds that can be used by practitioners. *Why it matters:* Decision thresholds in applications. *Existing limitation:* Asymptotic statements only. *Way forward:* Combine the paper's reduction with local laws for Hermitized matrices.

– **Gap:** Numerical study of the heavy-tailed regime. *Why it matters:* The slowest case sits at the boundary of the theorem. *Existing limitation:* No simulations beyond Bernoulli and Gaussian in the paper. *Way forward:* Systematic Monte Carlo over tail exponents, as in the reviewer's exploratory simulation above.

11 Research Opportunities

Idea 1. Finite-sample calibration of the circular law under fat tails

RESEARCH QUESTION. How does the finite-n error of the circular law depend on the tail index of the entry law, and can a tail-adjusted benchmark radius be derived?

MOTIVATION. The reviewer's simulation shows sup distance 0.114 at $n = 1200$ for a law with finite variance and no $2+\varepsilon$ moment, against 0.014 for Gaussian.

HYPOTHESIS. The error scales as the missing variance mass of the truncation at $n^{1/2}$-scale, of order $1/\log n$ for the borderline law and a power of n for tails with an exponent above two.

ECONOMIC MECHANISM. Heavy tails concentrate variance in a few entries that act as rank-one perturbations, shrinking the effective radius of the bulk.

LITERATURE CONNECTION. This paper; Bordenave and Chafai (2012); heavy-tailed random matrices.

DATA. Monte Carlo with Pareto-type, Student and log-corrected laws; optionally standardised daily returns of an index universe.

VARIABLES. Tail index, n, radial sup distance, effective radius.

METHOD. Simulation with regression of log error on $\log n$ per tail index; derive an analytic truncated variance correction.

IDENTIFICATION STRATEGY. Exogenous design of the entry law, so differences are causal by construction.

EXPECTED CONTRIBUTION. A table of finite-n corrections and a practical test with calibrated critical values.

MAIN RISK. Results may be a straightforward consequence of known extreme-value facts and add little beyond a computation.

Idea 2. Spectral null test for estimated VAR(1) matrices

RESEARCH QUESTION. Can the eigenvalue cloud of an estimated high-dimensional VAR(1) coefficient matrix be compared with the circular-law null to detect genuine lead-lag structure?

MOTIVATION. Estimating a 100×100 coefficient matrix from monthly returns produces a noise cloud that looks like a disk of radius $\sqrt{n/T}$.

HYPOTHESIS. Eigenvalues outside the disk of radius $\sqrt{n/T}(1 + o(1))$, after standardising residuals, indicate real dynamics, and their number is predictive of out-of-sample forecast gains.

ECONOMIC MECHANISM. Lead-lag relations between assets or sectors appear as outlier eigenvalues, noise as the bulk.

LITERATURE CONNECTION. Plerou et al. (2002) for the symmetric case; this paper for the null.

DATA. CRSP or Fama-French industry returns (not downloaded in this review; availability to be checked).

VARIABLES. Eigenvalue moduli, out-of-sample R^2 of the forecast with the outlier subspace.

METHOD. Ridge-estimated VAR(1), rolling windows, comparison with a bootstrapped null.

IDENTIFICATION STRATEGY. Placebo by cross-sectional shuffling of time labels, preserving marginal laws.

EXPECTED CONTRIBUTION. A rule for selecting the number of predictive modes in high-dimensional VARs.

MAIN RISK. Cross-correlation among assets breaks the iid null; one must pre-whiten with a factor model.

Idea 3. Interbank stability beyond the May threshold

RESEARCH QUESTION. When does the circular-law stability criterion of May (1972), as applied to interbank exposure matrices, fail because of non-zero mean, sparsity, or heterogeneity of exposures?

MOTIVATION. Corollary 1.15 shows the disk is robust to low-rank means, but real exposure matrices have a full-rank structure (core-periphery).

HYPOTHESIS. The spectral radius of the exposure matrix exceeds the disk prediction by an amount determined by the dominant eigenvalue of the mean block structure.

ECONOMIC MECHANISM. Contagion propagates along eigenvectors; a super-critical eigenvalue means amplification of losses.

LITERATURE CONNECTION. May (1972), Haldane and May (2011), this paper.

DATA. BIS locational banking statistics or ECB large exposures data, subject to access.

VARIABLES. Exposure matrix, capital buffers, spectral radius, DebtRank.

METHOD. Compare empirical spectra with block-structured random matrix predictions from Appendix C.

IDENTIFICATION STRATEGY. Cross-country variation in network density.

EXPECTED CONTRIBUTION. A theoretically anchored, data-calibrated stability indicator.

MAIN RISK. Data confidentiality and sparsity make the asymptotics inapplicable.

Idea 4. Heterogeneous variance profiles and sector-block matrices

RESEARCH QUESTION. What is the limiting support of $n^{-1/2}(M_n + C_n \circ X_n)$ for a block variance profile C_n, and how well does the Appendix C weak invariance principle describe finite matrices?

MOTIVATION. Equity sectors have different volatilities; the iid assumption is wrong at the level of variance.

HYPOTHESIS. The limiting measure is determined by a deterministic equation depending only on the variance profile, with support a disk only when the profile is doubly stochastic.

ECONOMIC MECHANISM. Sector-level heterogeneity in noise amplitude changes the stability threshold.

LITERATURE CONNECTION. Appendix C of this paper; Girko on variance profiles.

DATA. Simulation and sector-level volatility from public indices.

VARIABLES. Variance profile, support radius, outlier count.

METHOD. Numerical solution of the profile equation; comparison with eigenvalues of simulated matrices.

IDENTIFICATION STRATEGY. Controlled variation of the profile.

EXPECTED CONTRIBUTION. Concrete finite-n validation of an abstract theorem.

MAIN RISK. The profile equation is stated in the literature but not reproduced in this paper, so the derivation must be sourced independently.

Idea 5. Eigenvalue-based early warning from rolling lead-lag matrices

RESEARCH QUESTION. Does the excess of the leading eigenvalue modulus of a rolling estimated lead-lag matrix over its circular-law benchmark predict drawdowns in aggregate equity returns?

MOTIVATION. A spectral radius near one signals near-instability of the linear dynamics.

HYPOTHESIS. Excess spectral radius rises before crisis windows and predicts higher realised volatility at monthly horizons.

ECONOMIC MECHANISM. Increased cross-asset feedback precedes episodes of fire sales and contagion.

LITERATURE CONNECTION. Idea 2; Haldane and May (2011) on banking ecosystems; this paper.

DATA. Daily returns of 100 large US stocks, 1990 to 2026 (to be sourced).

VARIABLES. Excess radius, realised volatility, VIX, drawdown indicator.

METHOD. Rolling ridge VAR(1), predictive regressions with Newey-West errors, out-of-sample evaluation.

IDENTIFICATION STRATEGY. Predictive, not causal; protect against look-ahead by estimating only on past windows.

EXPECTED CONTRIBUTION. A new systemic-risk indicator with a theory-based benchmark.

MAIN RISK. Estimation noise and overlapping windows may produce spurious predictive power.

11.1 Ranking

Table 5. Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Fat-tail calibration	4	5	5	4	3	4	25
2. VAR(1) spectral null	4	4	5	4	4	4	25
3. Interbank stability	4	3	2	4	5	4	22
4. Variance profiles	3	4	4	4	3	3	21
5. Early warning indicator	3	3	4	3	4	3	20

Source: reviewer's subjective scoring.

The strongest three are ideas 1, 2 and 3. Idea 1 is the most feasible and directly follows from the one quantitative weakness found in this review. Idea 2 turns the theorem into a usable specification test with ample public data. Idea 3 has the highest importance but the weakest data access, so it ranks third despite the best economic motivation.

12 Learning Framework

12.1 Prerequisite concepts

- Weak and vague convergence of measures; convergence in probability versus almost surely; Borel-Cantelli lemmas.
- Singular values, the Schur decomposition, Weyl and Cauchy interlacing inequalities.
- Logarithmic potential theory in the plane and the Laplacian of $\log|z|$.
- Concentration of measure (Talagrand, Hoeffding, Chernoff) and the Marchenko-Pastur law.

12.2 Core concepts learned

- Hermitization: eigenvalues of a non-normal matrix are accessible through the singular values of the family $A - zI$.
- Universality is proved by replacement of a single functional, the log-determinant, rather than of the whole spectrum.
- Weak limits of empirical measures do not control extreme eigenvalues.
- The truncation level $n^{0.1}$ makes unbounded entries harmless.

12.3 Important equations

The log-potential identity (4), Girko's identity (6), the row-distance formula for the determinant (7), the lower-tail bound (8) with Talagrand's inequality (9), the negative second moment identity (10), the Dozier-Silverstein equation (12) and the Ginibre density (13).

12.4 Methods learned

Stieltjes-type and Fourier analysis of an ESD through the log-determinant; conditioning and truncation to remove unbounded entries; Borel-Cantelli with polynomially small tail events; interlacing to move between $n' \times n$ and $n \times n$ matrices; concentration for distances to subspaces.

12.5 Questions to test understanding

COMPREHENSION

1. State the circular law and explain why the normalisation $n^{-1/2}$ is needed.
2. What does condition (2) say about the mean matrix, and why is rank $o(n)$ enough in Corollary 1.15?
3. Why does the paper need Mehta's theorem even though it proves universality?

ANALYSIS

1. Derive Girko's identity (6) for a single eigenvalue.
2. Prove the negative second moment identity (10) and explain how it converts row distances into singular value control.
3. Why does the lower-tail bound only need a bound of the form $e^{-n^{0.01}}$ and not a sharper exponent?

RESEARCH

1. Can the replacement principle be proved with an explicit rate under a $2 + \varepsilon$ moment?
2. What replaces Talagrand's inequality when the rows are correlated through a factor structure?
3. How would one detect, from a single observed matrix of dimension 500, whether the entries are heavy tailed enough to bias the circular-law null?

12.6 Papers to read next

Tao and Vu (2008); Bordenave and Chafai (2012); Rudelson and Vershynin (2008); Gotze and Tikhomirov (2010); Chatterjee (2006); Sommers, Crisanti, Sompolinsky and Stein (1988).

13 Conclusion

13.1 One-sentence thesis

The eigenvalue distribution of a large matrix with iid entries of mean zero and unit variance is asymptotically uniform on the unit disk, because the normalised log-determinant, the one statistic that controls the eigenvalue measure, is universal under a second moment alone.

13.2 Three most important insights

- Non-normal eigenvalue problems can be solved by Hermitization together with a replacement principle that needs only the log-determinant to match.
- Rates of convergence for the Hermitized ESD are unnecessary, because the troublesome rows are controlled by a lower-tail distance bound and Talagrand concentration.
- The theorem is asymptotic and is silent on speed; the simulations here show it can be very slow near the boundary of the assumptions.

13.3 Strongest evidence

The proof itself, and its agreement with the numerical log-determinants in Table 2, where Gaussian and Bernoulli entries match the disk potential to within 0.004 at $n = 1000$.

13.4 Biggest weakness

No rate of convergence. For finite variance without a higher moment, the finite-n distance to the limit is large and shrinks only logarithmically.

13.5 Most important research gap

Universality for correlated entries and a quantitative, finite-n version of the law that practitioners can use.

13.6 Best research opportunity

Calibrating a fat-tail-adjusted circular-law null for estimated high-dimensional VAR matrices, then testing its out-of-sample value.

13.7 Three papers to read next

Tao and Vu (2008); Bordenave and Chafai (2012); Rudelson and Vershynin (2008).

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